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ScienceDirect
Acta Tropica 105 (2008) 67–73
ACTA TROPICA
www.elsevier.com/locate/actatropica

Abandoning small-scale fish farming in western Kenya leads to higher malaria vector abundance

Annabel F.V. Howard *, Francois X. Omlin

Human Health Department, International Centre of Insect Physiology and Ecology (ICIPE) P.O. Box 30772, 00100 Nairobi, Kenya

Received 1 November 2006; received in revised form 30 July 2007; accepted 29 September 2007
Available online 18 October 2007

Abstract

Fishponds become abandoned due to lack of access to both young fish and technical support and faster economic returns from other activities. Certain conditions found in abandoned fishponds, such as absence of fish and presence of aquatic vegetation, are conducive to the presence of malaria vectors. We conducted a district-wide fishpond census to determine the maintenance status and mosquito populations of fishponds in Kisii Central District in western Kenya. Two hundred and sixty one fishponds were found, 186 active (fish present) and 75 abandoned (fish absent). Vegetation was not significantly associated with the distribution of *Anopheles gambiae s.l.*, *Anopheles funestus* or culicines (Diptera: Culicidae) in active or abandoned ponds. The presence of fish, however, correlated significantly with the distribution of all mosquito species, with significantly higher mosquito densities in abandoned fishponds. *An. gambiae s.l.* was the most abundant mosquito species found in both active and abandoned ponds, being proportionally more abundant in the abandoned ponds. The proportion of *An. funestus* increased with altitude. Following the census the demand for fish to re-stock abandoned ponds rose by 67% when compared to the same time period in the previous year. This study highlights the potential public health problems associated with the abandonment of small-scale fish farming in the highlands of western Kenya.
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Keywords: Fishponds; Western Kenya; *Anopheles*; Man-made habitats; Mosquito larval ecology; Malaria

1. Introduction

In sub-Saharan Africa, where most of the global malaria burden is borne, the main vectors are *Anopheles gambiae s.l.* and *Anopheles funestus* Giles (Diptera: Culicidae). Although classically the larval habitats of *An. gambiae s.l.* are thought to be small, temporary, sunlit pools with algae (Gillies and Coetzee, 1987; Gimnig et al., 2001), these species of mosquito have also been found in relatively large, permanent, man-made water bodies in western Kenya (Lockhart et al., 1969; Carlson et al., 2004; Fillinger et al., 2004; Mutuku et al., 2006). Despite a global investigation into the relationship between large dams and malaria (Keiser et al., 2005), the link has only been investigated for smaller water bodies in a select few areas (Ripert and Raccurt, 1987; Ghebreyesus et al., 1999).

Small-scale fish farming has been continuously practiced in western Kenya for almost 50 years. Lockhart et al. (1969) found in 1959 that the 1000 ponds they surveyed in the whole of western Kenya were well maintained, but by 1961 they found many fish ponds abandoned with over-grown edges and "only given the minimum amount of maintenance". This abandonment they attributed to "bad farming methods" which were producing smaller than anticipated yields and because fish farming required more work than the farmers had expected. In addition, nowadays fishponds become abandoned for several other reasons, including the farmers switching their resources to invest in cash crops such as tea and coffee, demoralisation due to poaching and lack of extension services from the Fisheries Department (FD) (for example help with harvesting and advice concerning pond construction and maintenance). Also, when the farmers harvest the ponds they remove all the fish including the small fry, in order to sell or eat them, which means that ponds become 'abandoned' for a while even if the farmers want to carry on fish farming.

Fish farming is still a favoured community activity in western Kenya today with new ponds being constructed and stocked. It is promoted as a way to relieve the pressure on catches from Lake Victoria and as a source of income and protein for people living in rural areas (Government of Kenya, 2002; FAO Inland

* Corresponding author. Tel.: +254 727 801284; fax: +254 20 8632001.
E-mail address: afv.howard@gmail.com (A.F.V. Howard).

0001-706X/$ – see front matter © 2007 Elsevier B.V. All rights reserved.
doi:10.1016/j.actatropica.2007.09.010

Water Resources and Aquaculture Service, 2003). However, little attention is paid to possible dangers of abandoned fishponds in enhancing malaria transmission.

If a water body does not contain fish it is more likely to have mosquitoes than a water body with fish in (Petranka and Fakhoury, 1991; Fletcher et al., 1992) with higher mosquito numbers in the fish-absent when compared to the fish-present areas (Wu et al., 1991; Prasad et al., 1993; Ritchie and Laidlaw-Bell, 1994; Takagi et al., 1995). In addition it has recently been found that *An. gambiae s.l.* pupal occurrence is positively correlated with both habitat size (Minakawa et al., 2005) and habitat stability (Mutuku et al., 2006) in western Kenya. Most fishponds in western Kenya are large and contain water throughout the year and thus it would be expected that the pupal output from them would be very high.

Considering the above-mentioned issues there is a possible danger that fishponds without fish and where the vegetation has been allowed to grow could harbour large numbers of malaria vectors. Therefore, we conducted a district-wide census to identify the number and maintenance status of fishponds in the Kisii Central District in western Kenya. The numbers of mosquito immatures found in these fishponds was measured to obtain a better understanding about the role of these numerous large permanent water bodies in the production of disease vectors during the dry season.

2. Materials and methods

2.1. Study area

The census was carried out in the Kisii Central District of the Nyanza Province in western Kenya (Fig. 1). This rural district is a highland area divided into seven administrative divisions covering a total area of 649 km^2 with altitudes of 1400–2200 m above sea level. Rainfall averages over 1500 mm annually with two wet seasons (February to June and September to November). Mean annual maximum temperatures are 27 °C and 24 °C in the low and highland areas of the district respectively; the mean annual minimum temperatures are 16 °C and 14 °C in the low and highland areas, respectively (Government of Kenya, 2002). The population density in the district is one of the highest in Kenya with a mean of 750 people per km^2 reaching a maximum of over 1000 people per km^2 in some divisions. Malaria in this district is endemic but highly seasonal. *An. gambiae s.l.* and *An. funestus* are the main malaria vectors in Kisii Central District.

2.2. Fishpond census

A fishpond census was carried out in all divisions of Kisii Central District from the 5th January to the 6th February 2004 in collaboration with the Kenyan FD. The area including Kisii town (hereafter called Municipality) is part of Kiogoro, Mosocho and Suneka divisions but was treated separately.

The fishponds were classed as ‘active’ or ‘abandoned’ depending on the presence or absence of fish, respectively. The FD officer assessed whether the ponds contained fish by throwing a handful of fish food into the water and waiting to see whether fish rose to the surface, if no fish came to the surface after 10 min the pond was deemed to be without fish. The FD officer also visually identified the type of fish in active ponds as ‘tilapiine’, ‘catfish’ or ‘other’. In some cases the FD officer had recently had contact with the farmers and so already knew which fish type was in the pond. The ponds were further classified as well maintained (WM) if the banks were clear (Fig. 2a) and there was less than 10% vegetation coverage; not well maintained (NWM) indicated that the banks were not clear of vegetation (Fig. 2b) and/or vegetation coverage of the pond was more than 10%.

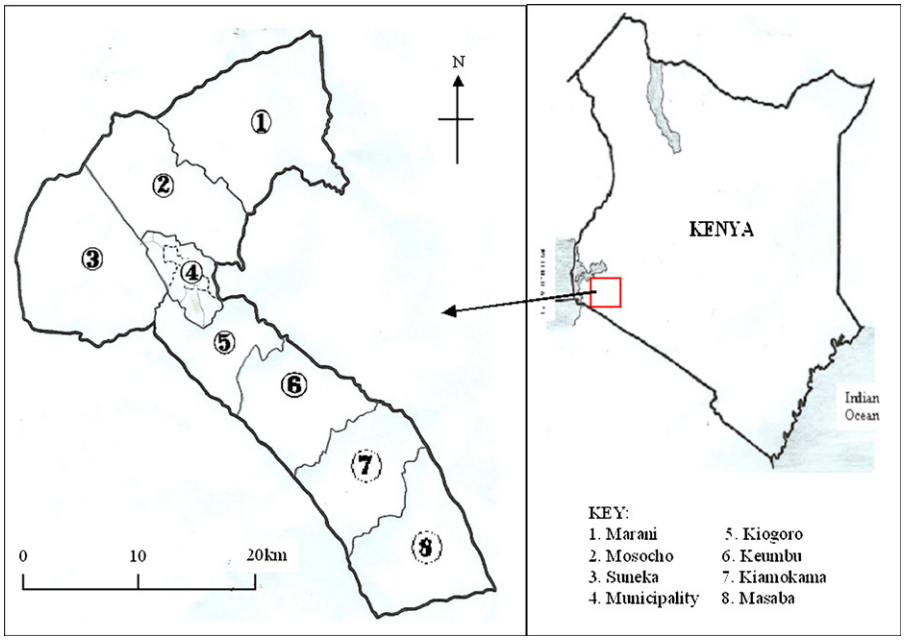


Fig. 1. Map of Kisii Central District and its location in Kenya.



Fig. 2. (a and b) Photographs showing A: the clear banks of a well maintained fishpond and B: the overgrown banks of a not well maintained fishpond.

Entomological assessments were carried out by taking five larval dips (2.5 L total volume) randomly from the periphery of each pond, with at least one dip from each side. Anophelines and culicines were distinguished and the anopheline mosquitoes were identified to species level using a morphological key (Gillies and Coetzee, 1987). The surface area of the ponds was measured and the owners of the ponds were educated about fishpond maintenance and the associated risk of mosquitoes breeding in abandoned fishponds.

2.3. Statistical analysis

After testing for equal/unequal variance, two-tailed Student's *t*-tests were carried out to investigate the differences between WM and NWM fishponds within the active and abandoned pond classes for each mosquito type. Two-tailed *t*-tests were also used to analyse any differences within each mosquito type between the active and abandoned pond types. Chi-squared tests were used to investigate the differences between the active and abandoned ponds with respect to the number of ponds containing mosquitoes of any type, *An. gambiae s.l.*, *An. funestus*, culicines, and the number of WM ponds. Within each pond type differences between the mosquito types were analysed using one-way ANOVA analysis. Statistical analysis was carried out at the 5% significance level using the SAS 9.1 software (SAS Institute Inc, 2004).

3. Results

3.1. Differences between WM and NWM ponds

Of the 186 active ponds found, 148 (80%) were WM, whereas just 13 (17%) of the 75 abandoned ponds were WM (Table 1); this difference was highly significant ($\chi^2 = 87.60$, d.f. = 1, $p < 0.0001$). The observed mean surface area of the WM and NWM ponds was not significantly different ($t = 1.10$, d.f. = 184, $p = 0.27$ for active; $t = 0.8$, d.f. = 73, $p = 0.42$ for abandoned ponds). Within the active ponds the proportion of *An. gambiae s.l.* in the NWM ponds decreases when compared to the WM ponds, with corresponding increases in the proportions of *An. funestus* and culicines. Within both the active and abandoned fishpond classes there were no significant differences for any of the mosquito types when comparing the WM and NWM ponds (Table 2). As such the active WM and NWM and the abandoned WM and NWM data was pooled for further analysis.

3.2. Differences between active and abandoned ponds

There was no significant difference between active and abandoned ponds with respect to the number of ponds containing mosquitoes of any type ($\chi^2 = 2.62$, d.f. = 1, $p = 0.11$). However, significantly more abandoned ponds contained *An. gambiae s.l.* ($\chi^2 = 13.83$, d.f. = 1, $p < 0.0002$), *An. funestus* ($\chi^2 = 5.68$, d.f. = 1, $p < 0.02$) and culicines ($\chi^2 = 6.65$, d.f. = 1, $p < 0.01$).

The proportion of *An. gambiae s.l.* increased in the abandoned ponds (Table 1). The mean number of *An. gambiae s.l.* found in the abandoned fishponds was also an order of magnitude higher than the mean number found in active fishponds and there were larger numbers of *An. funestus* and culicines in abandoned when compared to active fishponds. These differences were all statistically highly significant (Table 3), however, there was no significant difference between the mean surface

Table 1
Number (*N*) and surface area (mean ± S.E.) of the different pond types and the relative mosquito abundance (%)

Pond type	Vegetation	*N*	Area (m^2) mean ± S.E.	*An. gambiae s.l.*%	*An. funestus*%	Culicines%	Total%
Active	Well maintained	148	130.0 ± 7.58	54.7	28.7	16.6	100
Active	Not well maintained	38	148.9 ± 16.78	32.7	42.5	24.8	100
Active	Total	186	133.9 ± 6.94	48.1	32.9	19.0	100
Abandoned	Well maintained	13	118.2 ± 22.64	65.2	21.7	13.1	100
Abandoned	Not well maintained	62	143.2 ± 11.03	60.7	23.8	15.5	100
Abandoned	Total	75	138.9 ± 9.93	61.7	23.4	14.9	100

Table 2
Relative immature mosquito population densities in well maintained (WM) and not well maintained (NWM) fishponds

Pond type	Mosquito type	WM (mean ± S.E.)	NWM (mean ± S.E.)	*t*-stat	d.f.	*p*-Value
Active	*An. gambiae s.l.*	1.68 ± 0.21	1.66 ± 0.47	0.04	184	0.97
Active	*An. funestus*	0.88 ± 0.13	2.16 ± 0.64	1.96	40	0.06
Active	Culicines	0.51 ± 0.10	1.26 ± 0.59	1.28	39	0.21
Abandoned	*An. gambiae s.l.*	15.69 ± 3.03	11.18 ± 3.74	0.94	53	0.35
Abandoned	*An. funestus*	5.23 ± 3.77	4.39 ± 0.78	0.22	13	0.83
Abandoned	Culicines	3.15 ± 1.14	2.85 ± 0.92	0.20	30	0.84

areas of the active and abandoned fishponds ($t = 0.43$, d.f. = 259, $p = 0.67$).

3.3. Differences between mosquito types within the pond types

The ANOVA results showed a significant effect of the mosquito species on the number of mosquito immatures in both the active ($F_{2,555} = 8.92$, $p < 0.0002$) and abandoned ($F_{2,222} = 6.20$, $p < 0.003$) ponds. Therefore, Student–Newman–Keuls multiple comparison analysis was carried out (Table 3).

3.4. Fish type

In the 151 active fishponds (122 WM, 29 NWM) where the fish type was identified, 147 (97%) contained fish of the tilapiine subfamily (121 WM, 26 NWM). Of the WM ponds with tilapiine fish in, 27% had no mosquitoes in five dips compared to 12% of the NWM ponds containing tilapiine fish. However, this difference was not statistically significant using the Fisher's exact test ($p = 0.13$).

3.5. Levels of fishpond abandonment in the different divisions

A mean of 32 fishponds per division were found (23 active; 9 abandoned) (Table 4). In all divisions there were more active than abandoned fishponds. Suneka division had the highest rate of abandonment. There was no positive correlation between the percent active and abandoned ponds that were NWM within the divisions. Because the pond numbers in the different divisions varied widely and in some divisions were very low, no statistical tests were carried out between the different divisions.

3.6. Mosquito types per division

The proportion breakdown of the different mosquito types per division can be seen in Fig. 3a and b. In the abandoned ponds *An. gambiae s.l.* predominated especially in Municipality, Mosocho, Suneka and Marani, while in both the active and abandoned ponds the relative proportion of *An. funestus* was high in Keumbu, Masaba and Kiamokama and to a lesser extent in Kiogoro.

3.7. Community response to the census

Following the census the FD reported that the demand for both fish and help with renovating and restocking abandoned fishponds rose. For the first six months of 2004 there was a 67% increase in the demand for fish when compared to the same time period of 2003.

4. Discussion

As in previous studies (Petranka and Fakhoury, 1991; Fletcher et al., 1992) we found that more fish absent (abandoned) than fish present (active) fishponds contained mosquitoes,

Table 3
Relative immature mosquito population densities in active and abandoned fishponds after pooling the WM and NWM data

Mosquito type	Active Mean ± S.E.	Abandoned Mean ± S.E.	*t*-stat	d.f.	*p*-Value
An. gambiae s.l.	1.67 ± 0.19A[a]	11.96 ± 3.1A	3.27	75	<0.002
An. funestus	1.14 ± 0.17B	4.54 ± 0.90B	3.70	79	<0.0004
Culicines	0.66 ± 0.14C	2.90 ± 0.78B	2.83	79	<0.006

[a] The same letters in a column indicate no significant difference at the 0.05 level as calculated by Student–Newman–Keuls multiple comparison analysis, there is no relationship between the letters within the rows.

Table 4
Divisional breakdown of the number (*N*) of fishponds found and the proportion of active and abandoned fishponds not well maintained (%)

Division	Area (km^2)	Pond type			Not well maintained	
		Active	Abandoned	Total	Active	Abandoned
		N	*N* (%)	*N*	%	%
Municipality	29	18	8 (30.8)	26	38.9	75.0
Mosocho	105	10	7 (41.2)	17	20.0	57.1
Suneka	126	33	24 (42.1)	57	18.2	79.2
Marani	124	37	5 (11.9)	42	10.8	100.0
Kiogoro	61	46	11 (19.3)	57	17.4	90.9
Keumbu	71	13	8 (38.1)	21	23.1	100.0
Masaba	90	11	3 (21.4)	14	45.5	66.7
Kiamokama	72	18	9 (33.3)	27	16.7	88.9

although this difference was not significant when all the mosquito types were combined. However, when the mosquito types were separated, significantly more abandoned ponds contained mosquitoes of each type. This may be because *An. gambiae s.s.* avoids ovipositing in water that contains competitors (Munga et al., 2006). In this respect *An. funestus* may be seen as a competitor by *An. gambiae s.l.*, and it is also possible that a similar mechanism exists in *An. funestus*.

Our results indicate that fish significantly influence the distribution of anophelines and culicines in the study area. This phenomenon has also been found in previous studies in other geographical locations (Wu et al., 1991; Prasad et al., 1993; Ritchie and Laidlaw-Bell, 1994). Fish can influence mosquito numbers both directly and indirectly. Larvivorous fish directly

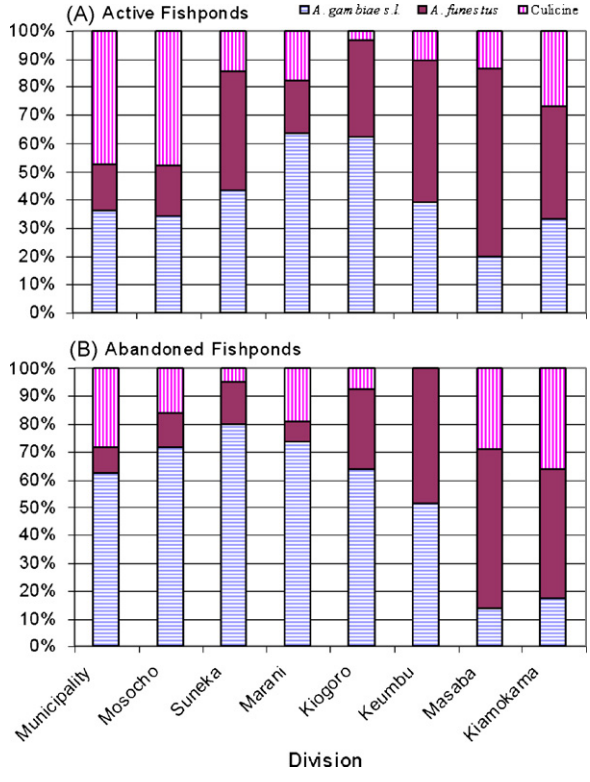


Fig. 3. (a and b) Proportional breakdown of mosquito types per division in A: active and B: abandoned fishponds.

control mosquito numbers through predation (Asimeng and Mutinga, 1993). In over 80% of the active fishponds the fish type was successfully identified, and in these cases, 97% of the fish belonged to the tilapiine sub-family. Although not identified in this study, the main tilapiine fish species farmed in the study area are *Oreochromis niloticus* (Perciformes: Cichlidae) and *Tilapia zilli* (Perciformes: Cichlidae). Under laboratory conditions both of these species have been shown to eat mosquito larvae, with one individual fish eating on average 300 mosquito larvae in just 15 min (Asimeng and Mutinga, 1993), while in the field *O. niloticus* has been shown to eat both mosquito eggs and larvae (el Safi et al., 1985). We have also demonstrated the ability of *O. niloticus* to control mosquito numbers in WM fishponds under field conditions in Kisii Central District; we found that the fish caused >94% reduction in malaria vectors in only 15 weeks (Howard et al., 2007). Fish indirectly influence mosquito numbers by causing ovipositing mosquitoes to selectively avoid fish containing water-bodies (Petranka and Fakhoury, 1991; Ritchie and Laidlaw-Bell, 1994; Angelon and Petranka, 2002). Although to our knowledge this has not been proven to occur in our study area.

Previous work has shown no relationship between vegetation and either *An. gambiae s.l.* (Minakawa et al., 1999) or culicines (Minakawa et al., 1999; Fillinger et al., 2004), and this was also found in this study. Despite the previously published positive association of *An. funestus* and aquatic vegetation (Lockhart et al., 1969; Gimnig et al., 2001) we found that vegetation did not significantly influence the distribution of *An. funestus* in this study.

Ndenga et al. (2006), in a study temporally contempory with ours, concluded that *An. gambiae s.s.* predominated in the highland areas while *An. funestus* predominated in a lowland area of western Kenya. They did, however, observe that in the lowland area there were more *An. funestus* favourable breeding habitats. Their highland area was Marani division (Kisii Central District) and they found 79.1% of the malaria vectors were *An. gambiae s.s.* (with all *An. gambiae s.l.* identified as *An. gambiae s.s.*). In Marani division we found *An. gambiae s.l.* constituted 63.9% and 73.4% of the active and abandoned fishpond mosquitoes respectively, and in both types of fishpond it was certainly the predominant mosquito species. Marani, however, is a relatively low-lying part of Kisii Central District (Government of Kenya,

2002). In Kiogoro, Keumbu, Masaba and Kiamokama, which are located at a higher altitude than the other divisions, we found the proportion of *An. funestus* to be relatively high which is likely to be an effect of altitude as this observation was made in both the active and abandoned fishponds. A study in Tanzania showed that *An. funestus* predominated in highland when compared to lowland areas throughout the year, with the reverse true for *An. gambiae s.l.* (Maxwell et al., 2003). This possible association of *An. funestus* with altitude is interesting and should be investigated further.

There was much variation in the rates of abandonment between the divisions. Kiamokama is rural and does not have good access to the market or FD extension services which may be why the rate of abandonment is higher than average. In contrast Kiogoro has easy access to the market and extension services and its rate of abandonment is below average. Municipality, which has both good access to the market and FD extension services also has a high level of abandonment, and this may be because in the town there are many other income generating opportunities which have a quicker return than fish-farming. Suneka, Mosocho and Keumbu have the highest rates of abandonment. In all of these three divisions some incentives were offered to start fish farming and this support has subsequently been withdrawn.

Kisii Central District was reported to have 554 fishponds in the year 2000 (Government of Kenya, 2002). The number of fishponds we found in this study was lower than anticipated possibly due to the transport problems. Kisii Central District is mainly a rural district with many areas inaccessible by vehicle. The topography is hilly and there are lots of trees so fishponds could not easily be visually identified. Since the census finished we have had much information regarding newly constructed and stocked fishponds, and about existing ponds that we missed during the census indicating that at present fishpond numbers in the district are much higher than reported here.

There were certain other limitations of this study, namely that it represents the situation at just one time point during the dry season. A comparative study during the wet season would be interesting to see if the dynamics of the different mosquito species changed at all, especially as we have found that mosquito numbers in fishponds decrease dramatically in the wet season (Howard et al., 2007). Also, information about the larval instar and pupal numbers would have been interesting to see if there was any difference between the active and abandoned ponds with respect to the instars present. Similarly, information about productivity of the fishponds in the form of adult mosquito emergence could have provided much information on the actual public health risk involved. Many of these issues were not undertaken due to the time constraints involved in trying to conduct a single time point census over such a large area where movement via roads is slow.

One of the main points from the 2005 Abuja Declaration on Sustainable Fisheries and Aquaculture in Africa is to develop aquaculture. Aquaculture worldwide has increased exponentially (FAO Inland Water Resources and Aquaculture Service, 2003; Keiser and Utzinger, 2005). Our findings indicate that aquaculture development should be accompanied by adequate public health education so that fish farmers know the potential risks of abandoning their ponds. The main problem at present is the lack of information and knowledge that the farmers have about these risks. We are working with the FD towards reactivating the abandoned fishponds in this area and the FD are educating the fish farmers they come into contact with about the dangers of abandoned ponds.

Our data indicates that abandoned fishponds could pose a public health risk in western Kenya, especially as the human population in Kisii Central District is around 750 people per km^2. As these fishponds contain water throughout the year this is potentially a year-round problem. This study suggests that for effective mosquito control and associated malaria prevention, all fishponds should be stocked with larvivorous fish. With new fishponds continually being constructed in the area we think that this is an important issue to address. Therefore we recommend the following:

- Public health officers and the FD should create wider awareness about the problem of abandoned fishponds among the fish farmers.
- Draining the water from abandoned ponds is not enough for mosquito control as in this area of western Kenya there is abundant rainfall that will gather in any depression in the ground. Therefore when a pond is to be abandoned it should be filled in.
- Fish should not be over-fed by their owners, as then they do not eat the mosquitoes and vegetation in the pond.

Acknowledgements

We would like to thank all the ICIPE-Kisii field staff for help with data collection, Zachary Kinaro and the other Fisheries Department officers involved, and Anthony Wanjoya for assistance with statistics. We would also like to thank two anonymous reviewers whose comments greatly improved the manuscript. This study was supported by the Government of Finland and BioVision.

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